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Regression Plots

Seaborn has many built-in capabilities for regression plots, however we won't really discuss regression until the machine learning section of the course, so we will only cover the **lplot()** function for now.

lplot allows you to display linear models, but it also conveniently allows you to split up those plots based off of features, as well as coloring the hue based off of features.

Let's explore how this works:

```
In [1]: import seaborn as sns
        %matplotlib inline
```

```
In [2]: tips = sns.load_dataset('tips')
```

```
In [3]: tips.head()
```

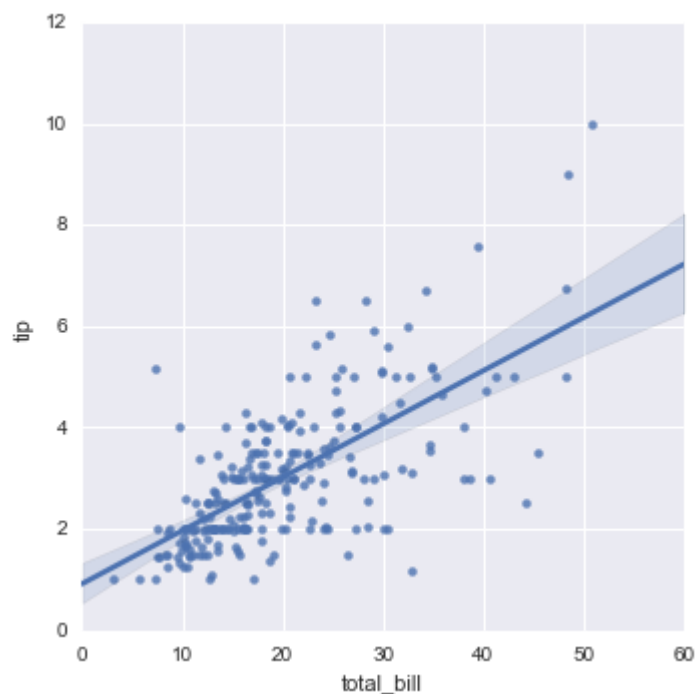
```
Out[3]:
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

lplot()

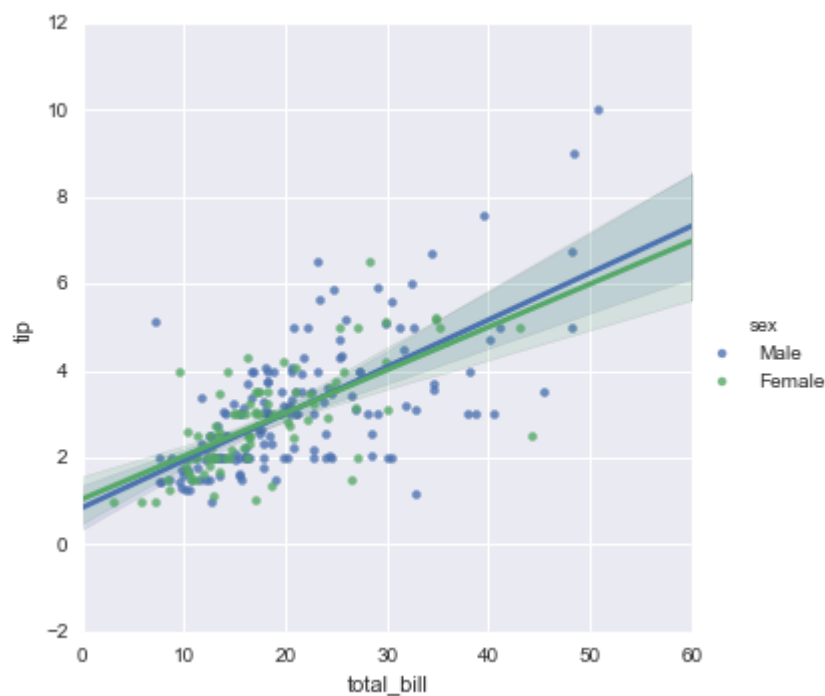
```
In [5]: sns.lmplot(x='total_bill',y='tip',data=tips)
```

```
Out[5]: <seaborn.axisgrid.FacetGrid at 0x117c81048>
```



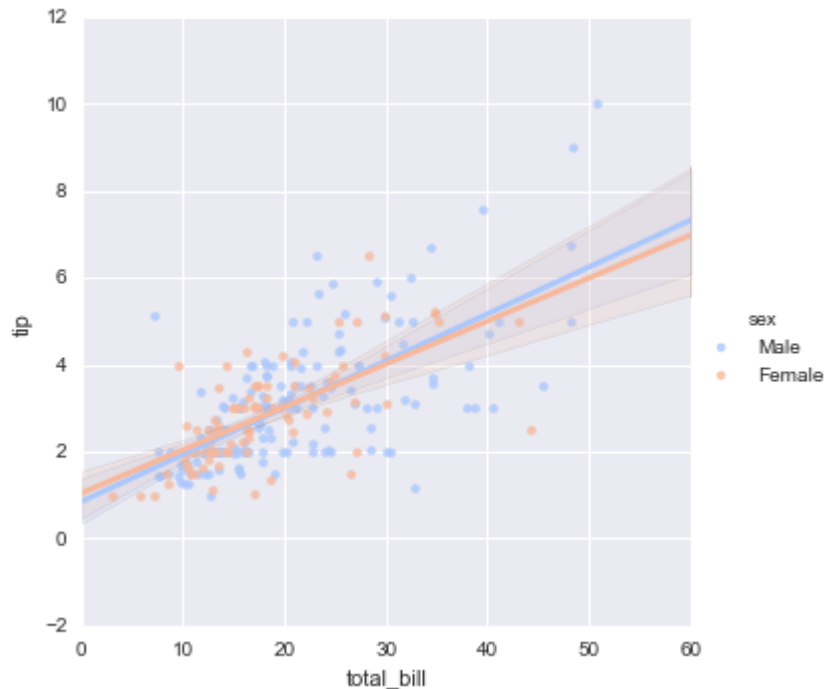
```
In [6]: sns.lmplot(x='total_bill',y='tip',data=tips,hue='sex')
```

```
Out[6]: <seaborn.axisgrid.FacetGrid at 0x11bf59f98>
```



```
In [13]: sns.lmplot(x='total_bill',y='tip',data=tips,hue='sex',palette='coolwarm')
```

```
Out[13]: <seaborn.axisgrid.FacetGrid at 0x11d0ca080>
```

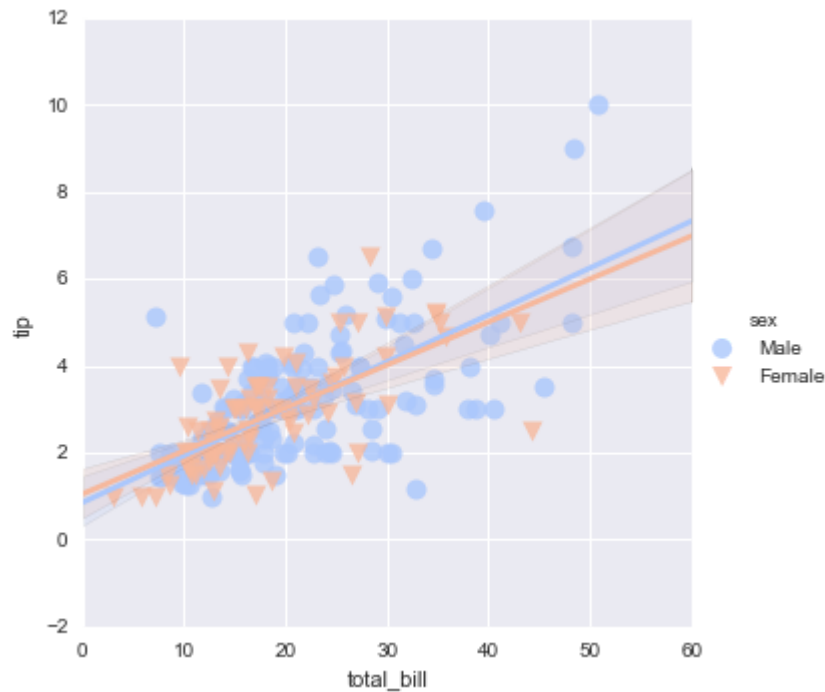


Working with Markers

`lmplot` kwargs get passed through to **regplot** which is a more general form of `lmplot()`. `regplot` has a `scatter_kws` parameter that gets passed to `plt.scatter`. So you want to set the `s` parameter in that dictionary, which corresponds (a bit confusingly) to the squared `markersize`. In other words you end up passing a dictionary with the base matplotlib arguments, in this case, `s` for size of a scatter plot. In general, you probably won't remember this off the top of your head, but instead reference the documentation.

```
In [16]: # http://matplotlib.org/api/markers\_api.html
sns.lmplot(x='total_bill',y='tip',data=tips,hue='sex',palette='coolwarm',
           markers=['o','v'],scatter_kws={'s':100})
```

Out[16]: <seaborn.axisgrid.FacetGrid at 0x11d8ab748>

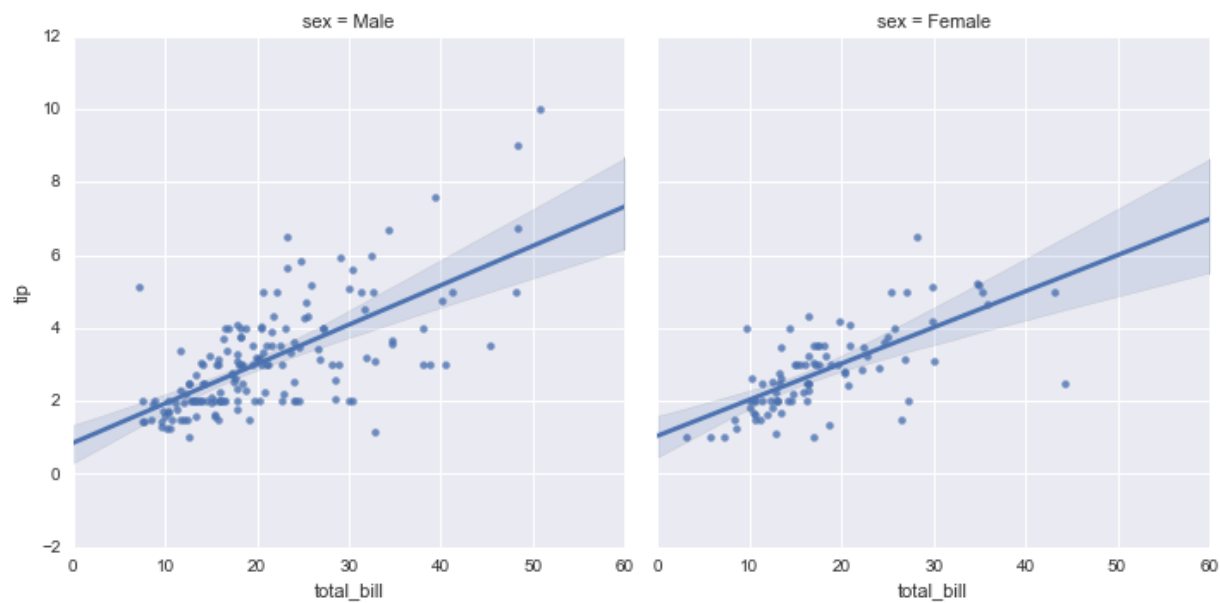


Using a Grid

We can add more variable separation through columns and rows with the use of a grid. Just indicate this with the col or row arguments:

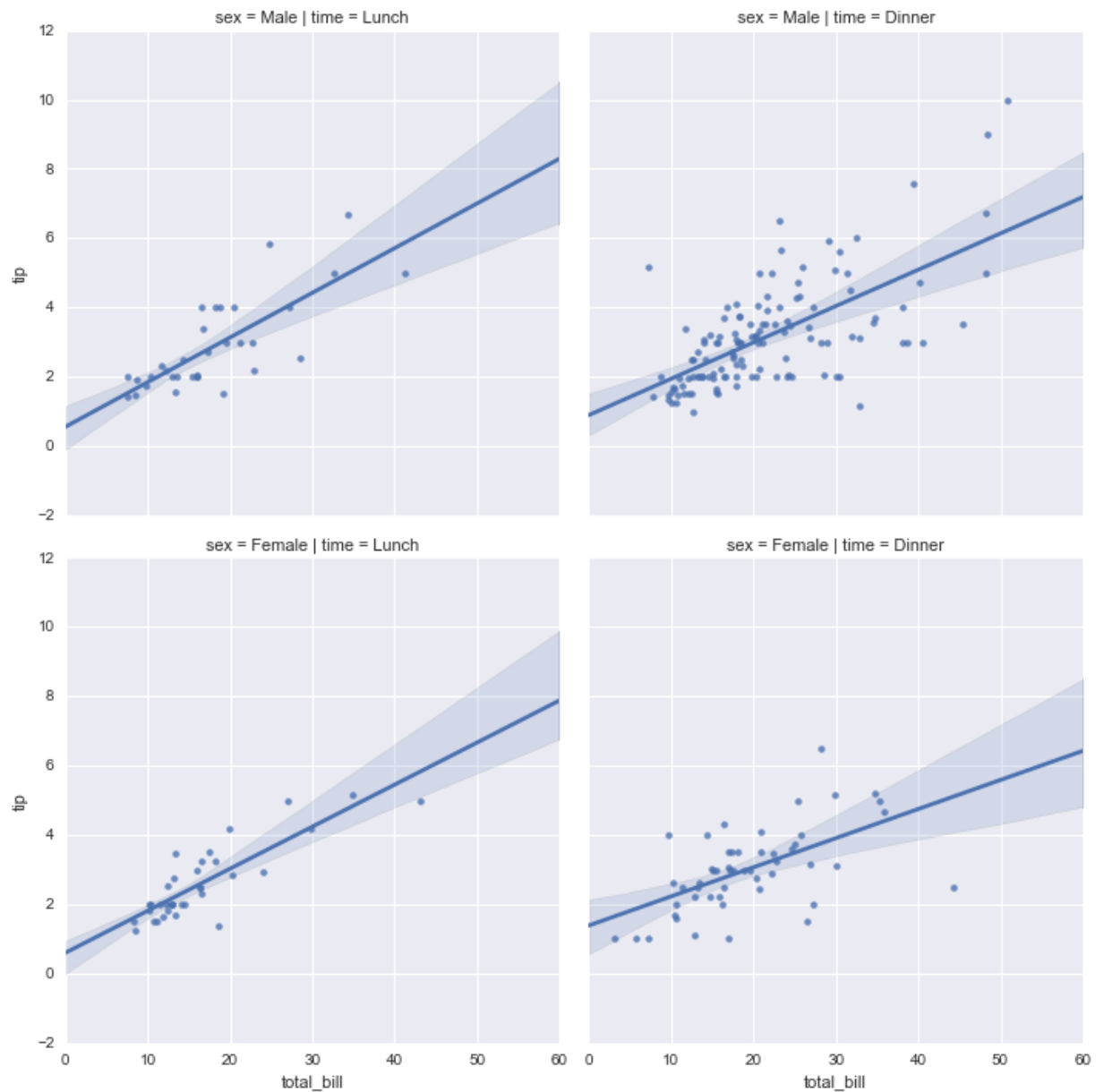
```
In [28]: sns.lmplot(x='total_bill',y='tip',data=tips,col='sex')
```

```
Out[28]: <seaborn.axisgrid.FacetGrid at 0x1215cd048>
```



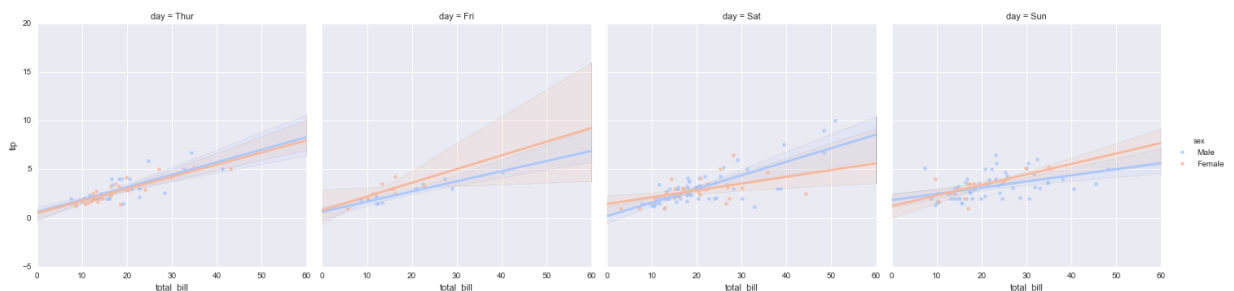
```
In [30]: sns.lmplot(x="total_bill", y="tip", row="sex", col="time", data=tips)
```

```
Out[30]: <seaborn.axisgrid.FacetGrid at 0x1226144e0>
```



```
In [24]: sns.lmplot(x='total_bill', y='tip', data=tips, col='day', hue='sex', palette='coolwarm')
```

```
Out[24]: <seaborn.axisgrid.FacetGrid at 0x11cbb02e8>
```

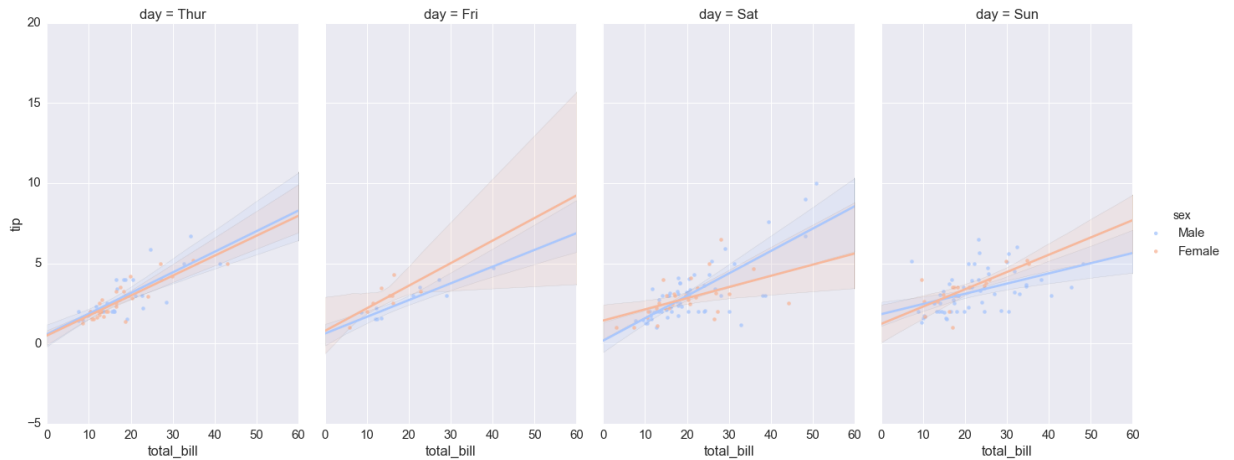


Aspect and Size

Seaborn figures can have their size and aspect ratio adjusted with the **size** and **aspect** parameters:

```
In [36]: sns.lmplot(x='total_bill',y='tip',data=tips,col='day',hue='sex',palette='coolwarm',  
                  aspect=0.6,size=8)
```

```
Out[36]: <seaborn.axisgrid.FacetGrid at 0x12420e5c0>
```



You're probably wondering how to change the font size or control the aesthetics even more, check out the Style and Color Lecture and Notebook for more info on that!

Great Job!